

Algebra 1
Unit 1
Lesson 6 Practice

Name _____
Date _____
Period _____

1. Explain mathematically why the expressions $\left(a^{\frac{1}{c}}\right)^b$ is equivalent to $(a^b)^{\frac{1}{c}}$ using the expressions $\left(4^{\frac{1}{2}}\right)^5$ and $(4^5)^{\frac{1}{2}}$.

2. Identify all of the expressions equivalent to $\left(49^{\frac{1}{2}}\right)^5$.

- $49^{\frac{2}{5}}$
- $(49^5)^{\frac{1}{2}}$
- $\sqrt[5]{49^2}$
- $49^{\frac{5}{2}}$
- $(\sqrt{49})^5$

For questions 3-7, complete each row with equivalent expressions.

	$a^{\frac{b}{c}}$	$\left(a^{\frac{1}{c}}\right)^b$	$(a^b)^{\frac{1}{c}}$	$(\sqrt[c]{a})^b$	$\sqrt[c]{a^b}$
3.		$\left(256^{\frac{1}{4}}\right)^3$			
4.					$\sqrt{64^3}$
5.	$128^{\frac{5}{7}}$				
6.			$(-27^2)^{\frac{1}{3}}$		
7.				$(\sqrt{81})^3$	

For questions 8-10, rewrite each expression using a rational exponent in the form $(a)^{\frac{b}{c}}$.

8. $\sqrt[3]{(-44)^5}$

9. $(\sqrt{289})^3$

10. $\frac{1}{\sqrt[5]{30^2}}$